



AIFARMS

Artificial Intelligence for Future Agricultural
Resilience, Management, and Sustainability



Enhancing Agricultural Research and Outreach at Tuskegee University

*Integrating Autonomous Crop Phenotyping Tools and Expanding
Outreach Efforts*

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Our Team



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Objective 1: Enhancing Research Capacity

Goal: Strengthening research infrastructure using cutting-edge autonomous technologies to enhance agricultural data collection.



**FarmBot
TerraSentia Rover
Amiga (Farm-ng)
Multispectral Drones
TubeBot
FarmBot + TubeBot Combination**

The Greenhouse

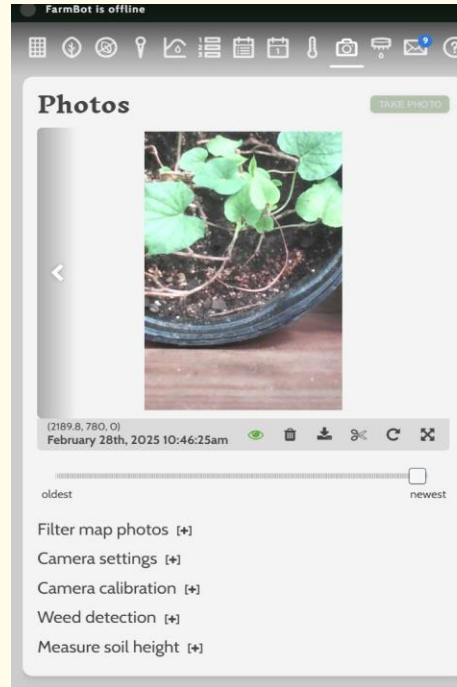
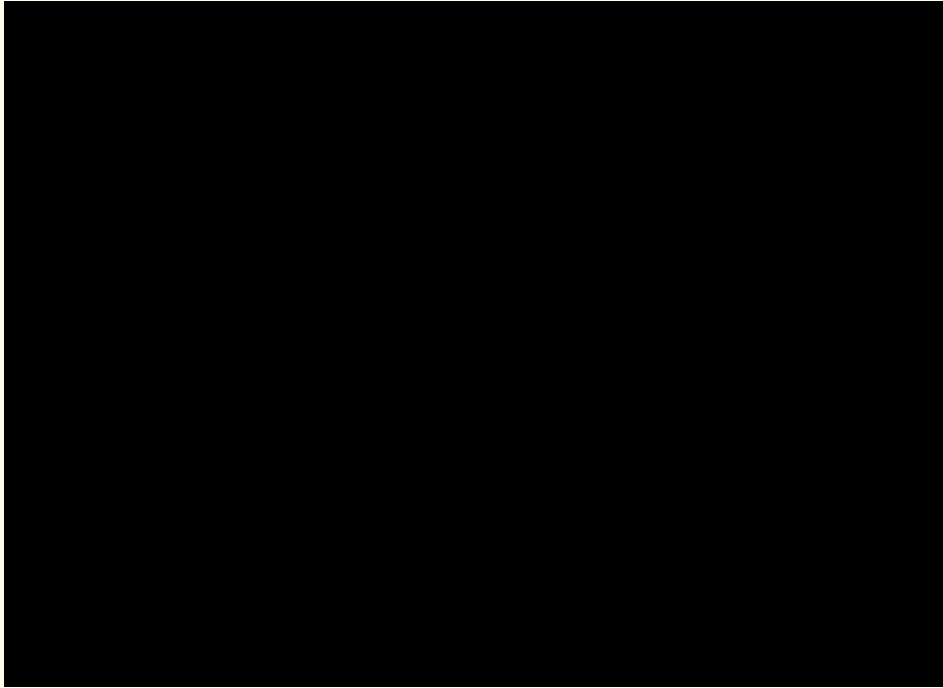


Tomatoes and
Sweetpotatoes

We utilize our greenhouse to monitor plant growth, collect chlorophyll data, and test AI-driven solutions in a controlled environment before deploying them in the field.



FarmBot



Open-Source:

Hardware designs and software are freely available. Easily modified for special use cases.

Automation:

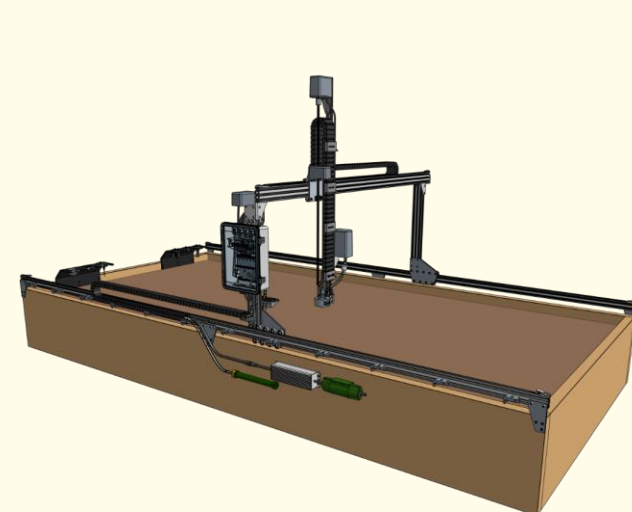
Using the block and LUA programming, we can create planting, watering, and weeding with high accuracy.

Web Interface:

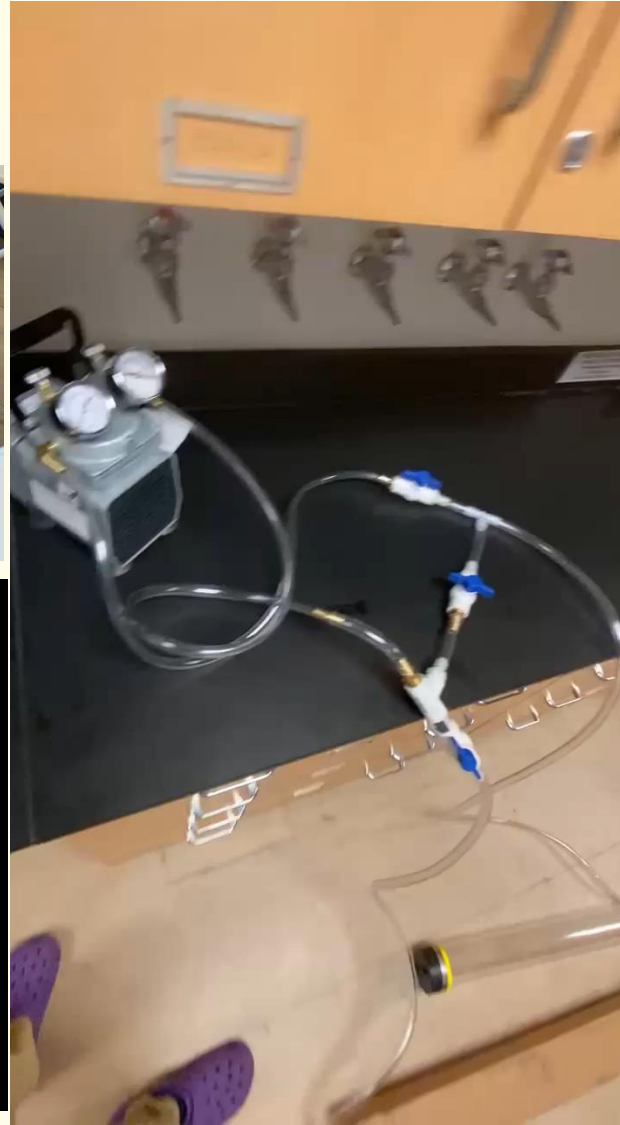
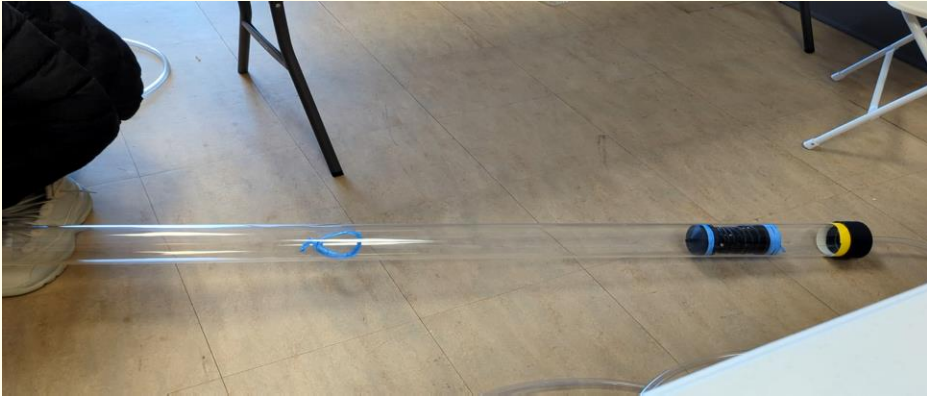
FarmBot can be controlled and programed from anywhere, on any device connected to internet.

Enviromental Savings:

Each plant is cared for with the ideal amount of water, reducing waste



TubeBot



Purpose:

Monitor root growth and behavior without disturbing the plant.

Design:

A clear buried tube allows for the robot to move along it with vacuum pressure. Has both control and experiment planters.

Features:

Bi-Directional movement / lights for image quality / wireless video / extendable tube length / multiple tests with one camera

Applications:

Root development, and stress responses. Crop improvements and soil interactions. Education visualization and environmental monitoring.

Developed by Dr. Rob Shepherd Lab, Cornell, Dr. Anand Mishra, a postdoctoral researcher in Shepherd's lab, who assisted with the virtual assembly and shipping of parts; Dr. Taryn L. Bauerle, who contributed to the development of the TubeBot. Designed for below-ground phenotyping, with a focus on root and tuber analysis.

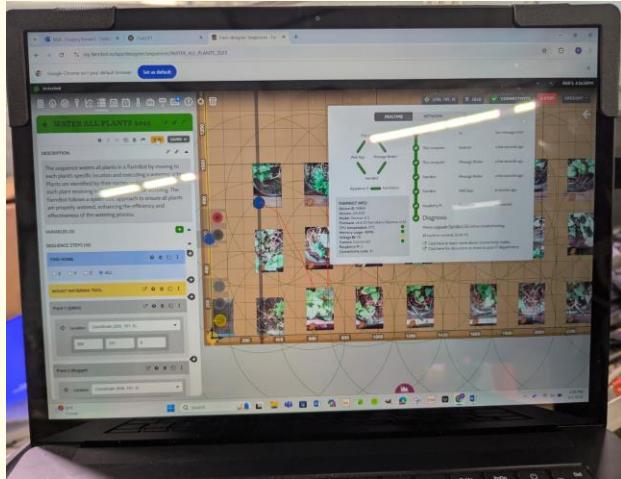
Automatous Crop Phenotyping System



- **FarmBot:**
Plant height, leaf health, and canopy, while autonomously watering and weeding
- **TubeBot:**
Underground root images, captures root growth, plant health, and soil interactions
- **FarmBot and TubeBot Combo:**
Above and below ground phenotyping allows for simultaneous observation of plant performance
- **Phenotyping:**
Something about the benefits of the phenotyping or how you can get hints about how the plant is growing



FarmBot, TubeBot

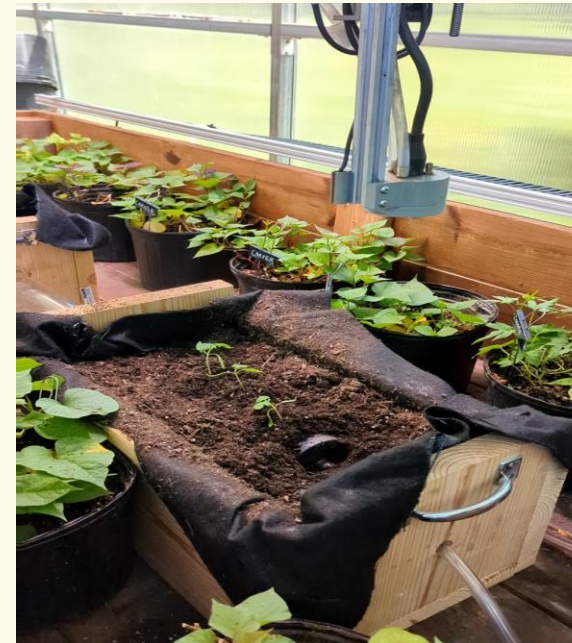


FarmBot Simulation



TubeBot Frame
Constructed by Mr. Daniels

TubeBot after potting



TubeBot and FarmBot



Autonomous Tools Integration



Amiga Tractor Robot

The Amiga robot will be used to manipulate fields, weeding, watering, plowing, sewing, as well as perform crop analysis



Multispectral Drones

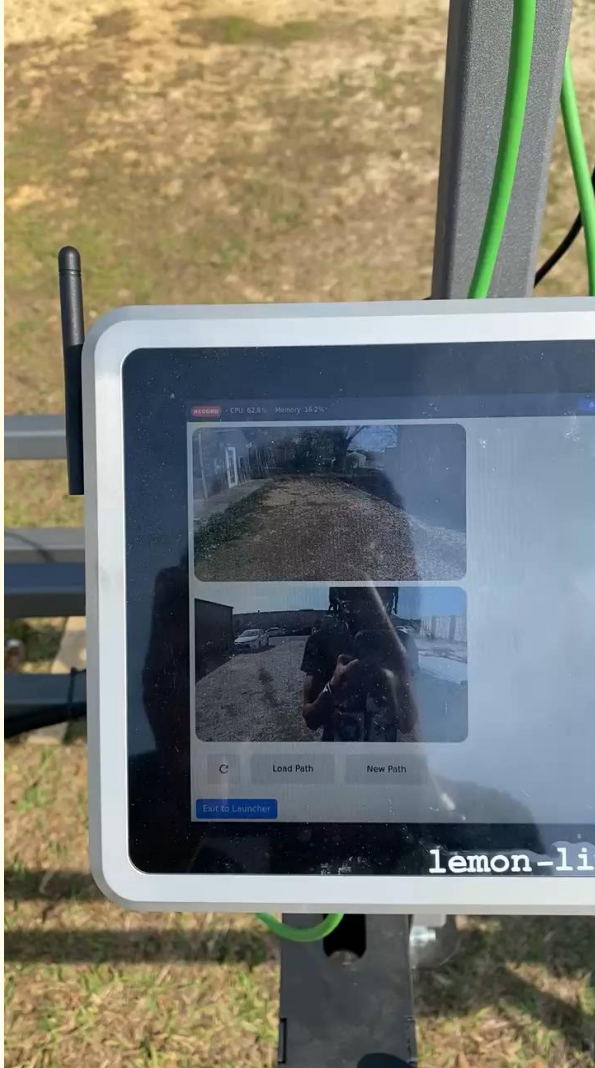
We utilize our drones in combination with preplaced flags to take dozens of photos that will later be processed in a vegetation index to determine crop health



EarthSense TerraSentia Rovers

Rovers capture images of the root crowns to create phenotype data, giving farmers key insights into field health and identifying potential concerns.

Amiga (Farm-ng) - Field Phenotyping



Intelligence Kit AI Brain:

Real Time Data Processing and Plant Recognition

Autonomous Navigation:
Uses GPS, LiDAR, and cameras to navigate crop rows without human input

Crop Phenotyping Support:
Captures plant height, spacing, leaf area, and health indicators

ROS Framework:
Allows for customization of data, custom processes, and integration with phenotyping models.



EarthSense TerraSentia Rover



Small & Rugged:

Compact all-terrain field robot designed for plant phenotyping.

Easily Controlled:

Uses GPS and cameras to navigate crop rows with minimal input

Intelligent Data Collection for phenotyping:
Collects stem width, leaf area, crop height, leaf color and unique pictures.



Capacity Building: Research Aerial Crop Monitoring Using UAVs

Unmanned Aerial Vehicles(UAV) equipped with multispectral cameras.

- Applications:
 - Large-scale monitoring of crop health.
 - Stress detection (e.g., water, nutrient deficiencies).
 - Mapping and yield estimation.
- Efficient means relative to "walking the back 40'(acres)"





Objective 2: Research and Outreach Impact

Research Impact: (Hooks Farm):

Data-Driven Insights: Crop health data collection enhances the precision and efficiency of agricultural research.

Technological Advancement:

Integration of autonomous systems enables more scalable and effective phenotyping.

Community Outreach Impact:

Local farmers are empowered with knowledge and hands-on experience with new agricultural technologies.

Student Development:

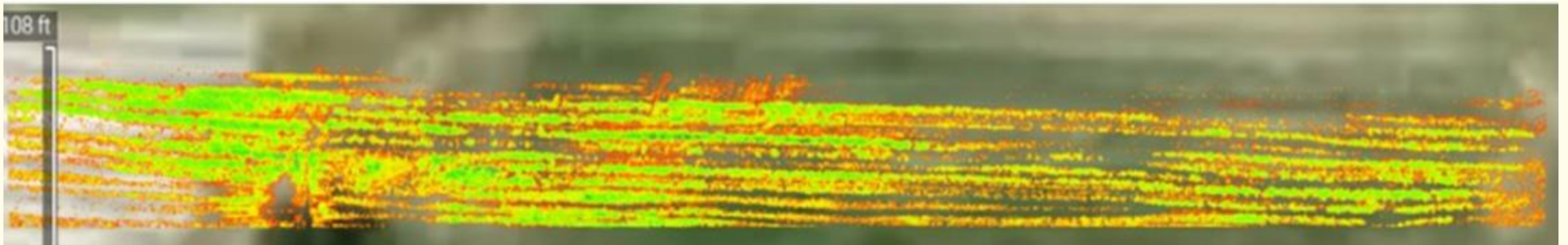
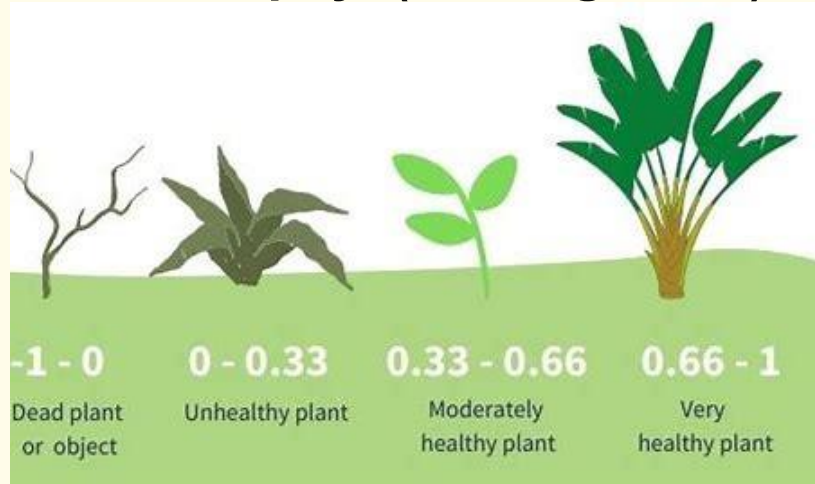
K-12 students and university students are exposed to cutting-edge tools, methodologies and possible careers.



NDVI Analysis:

Multispectral Drones (which utilizing RGB spectral imaging) used for the analysis of vegetative indices such as Normalized Difference Vegetation Index(NDVI).

Green = More chlorophyll (better growth)



Vegetation Reflectance



Vegetation indices with percentage of radiations emitted

The **vegetation indices** are a **combination of the percentage of refracted radiation** in different specific bands.

HEALTHY
VEGETATION REFLECTANCE

50% NIR 8% RED



NDVI = 0.72

STRESSED
VEGETATION REFLECTANCE

40% NIR 30% RED



NDVI = 0.14

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}$$



Overview of Tuskegee University's Vision in Agricultural Research

At Tuskegee, we are committed to advancing agricultural research by integrating state-of-the-art technologies to improve crop production and health.

Key Focus Areas:

- **Research Capacity:**
 - Leveraging cutting-edge autonomous tools for crop phenotyping and data collection
- **Outreach:**
 - Engaging and training a diverse group of stakeholders including K-12 students, farmers, educators, and government officials, to apply these technologies.



Hooks Farm

Coverage at a local farm that produces
Strawberries, Collards, Rutabaga and Turnips.

Readings Taken:

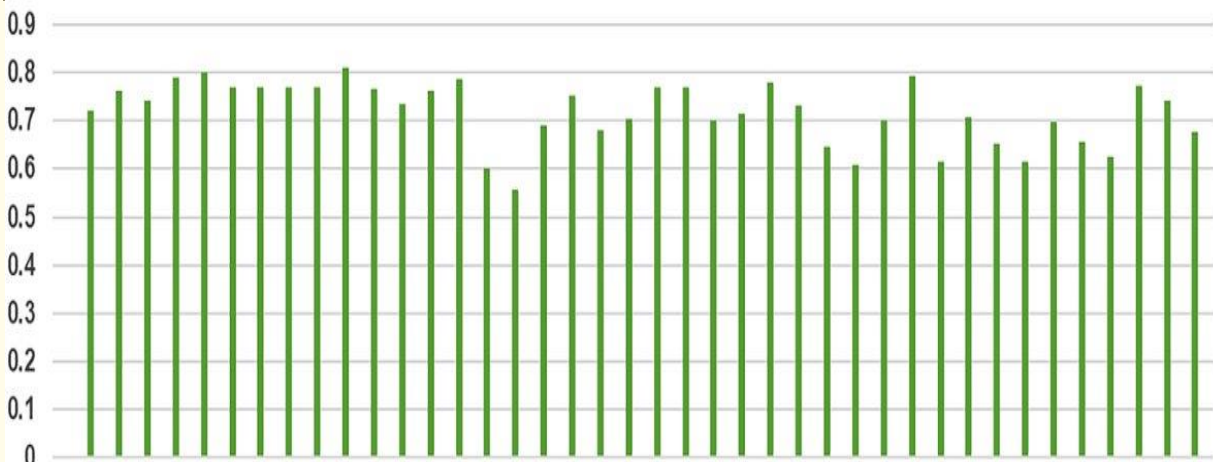
- Avg Chlorophyll
- Avg plant height
- Avg NDVI

Key Observations:

- High chlorophyll content and NDVI values indicate strong plant health

Recommendation:

- Spacing Adjustments
- Soil Testing



Annotation



Areas Of Future Exploration



Technology:

Creating Amiga attachments for weeding, plowing and other processes.

Test Farm/TubeBot collaboration

Larger Outreach:

Expand outreach events to more farmers, student, and officials.

Provide free crop health assessments, and tech use

Collaboration:

Build further partnerships with universities, research labs, and agricultural organizations.



Impact



Crop Quality:
Boosts crop yields with precise data from NDVI and chlorophyll analysis

Larger Outreach:
Supports farmers by automating scouting and reducing manual fieldwork

Youth Engagement:
Increasing the interest in Ag in younger people

Crop Insights:
Improves crop health monitoring through multispectral imaging and drone mapping

Conclusion



AIFARMS

Artificial Intelligence for Future Agricultural
Resilience, Management, and Sustainability



Tuskegee University - Through our innovative approach, we are strengthening both research capacity and community involvement, ensuring the sustainable growth of agriculture.

Future Goals:

- **Expand Tool Integration**
- **Broaden Our Reach**
- **Develop Collaborations**

The Impact on the team?

